

Zahra Amini

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🐙 github.com/aminizahra

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Education

B.Sc. in Software Engineering

2015 – 2020

Sirjan University of Technology

Sirjan, Iran

- GPA: 14.61/20 (German Grade: 2.6) - Major GPA: 16.747 (German Grade: 1.97)
- Thesis: Large-scale Semantic Point Cloud Segmentation with Superpoint Graphs (Grade: 20/20)

High School & Pre-University in Mathematics and Physics

Sep 2011 – Jun 2015

Farzanegan (Exceptional Talents) High School

Shahrbabak, Iran

- Graduated from the highly selective National Organization for Development of Exceptional Talents

Notable Courses

- **Artificial Intelligence & Data:** Fundamentals of Computational Intelligence, Bachelor's Degree Project, Principles of Robotics, Fundamentals of Language and Speech Processing, Fundamentals of Computer Vision, Database, Data Structure, Algorithms Design.
- **Mathematics & Statistics:** Engineering Statistics and Probability, Signals and Systems, Discrete Mathematics, Engineering Mathematics, Differential Equations, General Mathematics 1 & 2.
- **Programming & Systems:** Fundamentals of Secure Computing, Systems Analysis & Design, Principles of Compiler Design, Embedded and Real-Time Systems, Fundamentals of Computer and Programming, Programming Languages Design, Advanced Programming.

Research Interest

Biomedical Data Analytics & Time-Series Signal Processing | Explainable AI (XAI) & Clinical Decision Support Systems | Advanced Computer Vision & Medical Image Semantic Segmentation | Industrial Anomaly Detection & Predictive Maintenance

Publications

- [1] Seyed-Ali Sadegh-Zadeh, Mahshid Bagheri, Zahra Amini, Shayan Saadat, Mohammad Amin Barati, Delaram Jarchi, "A calibrated prediction model for general anaesthesia triage in paediatric dentistry: development, explainability, and preliminary external evaluation." Under Review

Developed a calibrated ML prediction model utilizing clinical tabular datasets for general anesthesia triage, applying advanced explainable AI (XAI) techniques to ensure clinical transparency.

- [2] Zahra Amini (Lead Author), et al., "Introduction to Artificial Intelligence in Healthcare," In Press
Shahid Beheshti University of Medical Sciences Press.

Authored an interdisciplinary textbook bridging machine learning, data analytics, and healthcare, designed to provide foundational and practical AI knowledge for medical students and professionals.

- [3] Zahra Amini, "Large-scale Semantic Point Cloud Segmentation with Superpoint Graphs," Spring 2020
B.Sc. Thesis, Dept. of Software Engineering, Sirjan University of Technology.

Developed a comprehensive 3D point cloud processing framework for semantic segmentation under the supervision of Dr. Amir Salarpour, achieving a perfect score for exceptional academic execution.

Research Experience

Research Assistant

Feb 2024 – Present

Supervisor: [Dr. Seyed-Ali Sadegh-Zadeh](#)

University of Staffordshire

- **Pediatric Dentistry Triage (Healthcare Data Analytics)**
 - Developed and validated a calibrated machine learning prediction model utilizing clinical tabular datasets for general anesthesia triage in pediatric dentistry
 - Applied advanced explainable AI (XAI) techniques to interpret model decisions and ensure clinical transparency

– Synthesized research outcomes into a co-authored manuscript, currently under external review

Data Science Research Consultant

Jun 2024 – Nov 2024

National Iranian Copper Industries Co. (NICICO)

- Developed predictive maintenance and anomaly detection models for heavy industrial machinery using time-series sensor data and Isolation Forests
- Conducted statistical data analysis and designed automated ETL pipelines to optimize resource allocation and operational efficiency

Research Assistant & Undergraduate Thesis Student

Feb 2019 – Feb 2021

Supervisor: [Dr. Amir Salarpour](#) [[View Recommendation](#)]

Sirjan University of Technology

- Engineered a scalable machine learning framework for large-scale 3D point cloud semantic segmentation utilizing Superpoint Graphs
- Demonstrated deep technical mastery of artificial intelligence and advanced fundamental concepts in Speech and Language Processing

Academic Teaching Experience

Undergraduate Teaching Assistant

Jul 2018 – Mar 2022

- Data Structures (Jul 2018 – Mar 2022) [[View Recommendation](#)] Sirjan University of Technology
- Basics & Advanced Programming (Sep 2019 – Sep 2020) [[View Recommendation](#)]
- Algorithm Design (Sep 2018 – Sep 2020) [[View Recommendation](#)]

Teaching Experience

Technical Content Developer (AI Fields)

Feb 2023 – Present

[Hobot Academy](#) (Persian) & [AlwithZahra](#) (English) (YouTube Platform)

- Designed and recorded foundational-to-advanced AI courses and roadmaps videos
- Reviewed and translated complex academic papers
- Hosted live technical interviews with industry experts

Instructor

Mar 2025 – Dec 2025

National Artificial Intelligence Association of Iran

- Instructed the "Friday AI Series" workshops, covering **Medical Image Analysis** (U-Net, LV data), **Tabular Data Preparation**, and **Numerical Data Modeling** for price prediction. [[GitHub Repo](#)]

Senior Instructor & Lead Author

Feb 2025 – May 2025

Course: Neuro-AI

Student Scientific Association of Neurosurgery

- Architected interdisciplinary ML and data analytics curriculum for 200+ medical students
- Taught and supervised end-to-end practical projects focusing on healthcare AI and clinical dataset analysis in Python
- Directed course outcomes into a co-authored textbook: "Introduction to Artificial Intelligence in Healthcare" (In Press, Shahid Beheshti University of Medical Sciences Press)

Instructor

Oct 2024 – Nov 2025

Course: Python for ML, Applied Mathematics for AI, and Version Control(Git/GitHub)

Hobot Academy

- **Instructed AI Mathematics:** Linear Algebra (SVD), Probability (Hypothesis Testing), and Optimization (Gradient descent).
- **Taught Information Theory:** Entropy, KL divergence, and PCA for dimensionality reduction.
- **Guided Python & Git:** Data manipulation (NumPy, Pandas, Seaborn), Regex, and collaborative version control workflows on GitHub.

Senior Instructor

Sep 2024 – Dec 2024

Course: Comprehensive Machine Learning [[View Recommendation](#)]

SmartEra Organization

- Delivered technical instruction covering the mathematical foundations of Machine Learning, statistical hypothesis testing, EDA, and predictive modeling (supervised, unsupervised, and ensemble/boosting algorithms)

Instructor

Sep 2022 – Dec 2024

Courses: Hands-On Machine Learning (Scholarship), Hands-On Deep Learning (Scholarship), ML at NLP School

Filogor

- [[View Work Certificate](#)] | [[NLP Certificate](#)] | [[DL Certificate](#)]

- Engineered and taught comprehensive syllabi spanning the data analytics lifecycle, from EDA (Pandas, NumPy) to implementing ML algorithms (Statistical tests, Regression, Random Forest, SVM, PCA, DBSCAN) both from scratch and via scikit-learn.
- Delivered advanced deep learning architectures (CNNs, GANs, RNNs, LSTMs, GRUs, Transformers, ViT) using Keras, PyTorch, and TensorFlow, with a strong emphasis on their mathematical foundations.
- Directed the curriculum roadmap for the “ML for NLP” program, instructing practical text analytics and guiding students through regex-based preprocessing, feature extraction (TF-IDF), and word embeddings (Word2Vec, GloVe) for text classification.
- Spearheaded the Pro Bono AI Scholarship initiative, democratizing education by mentoring 200+ students to deploy real-world models (e.g., recommendation systems, Medical Image Segmentation) and manage collaborative workflows using Git and GitHub. [\[Student Reviews\]](#)

Teaching Assistant

Oct 2020 – Sep 2024

[\[View Recommendation\]](#) | [\[View Work Certificate\]](#)

Filogier

- Assisted in 5 bootcamps including Data Science, Computer Vision, and Data Engineering (Jan 2022 – Sep 2024); designed problem-solving sessions, evaluated assignments, and developed educational resources.

Data Science Mentor

Jul 2023 – Sep 2023

KaggleXBIPOC Mentorship Program [\[View Certificate\]](#)

Kaggle

Professional Experience

Founder

Aug 2024 – Present

Hobot Academy

- Founded and developed the academy, focusing on foundational and advanced AI education. [\[Visit Website\]](#)

Director of Training

Jan 2025 – Dec 2025

Directed training programs, specialized AI courses, and educational strategies.

SmartEra Organization

Data Scientist

Jan 2025 – Jun 2025

National Iranian Copper Industries Co. (NICICO)

- Developed a specialized Large Language Model (LLM) assistant for cross-departmental technical experts, integrating academic research papers and proprietary corporate documents as internal knowledge bases.
- Analyzed mining blast data to optimize ore selection, identifying high-yield extraction sources to maximize copper purity.

Head of Data Education Department

Jan 2023 – Dec 2023

- Managed the Data Science School, directed the Mathematics Scholarship Course, and oversaw curriculum planning and program execution.

Filogier

Database Engineer & Data Engineering Intern

Apr 2018 – Oct 2019

Gol Gohar Iron and Steel Development Co. (GISDCO)

- **Database Engineer** (Jun 2018 – Oct 2019)
 - Designed and architected the organizational database in **SQL Server**, managing schema design, relational table creation, and database structures.
 - Wrote and optimized complex SQL queries for efficient data retrieval, manipulation, and reporting.
- **Data Engineering Intern (HR Systems)** (Apr 2018 – May 2018) | [\[View Certificate\]](#)
 - Completed a comprehensive 320-hour internship program.
 - Managed organizational databases using Excel, applied formulas, and generated reports.
 - Operated web-based organizational software (Kasra) and assisted in preparing analytical reports to adjust the company's organizational chart.

Skills

Technical: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Science, Bio-Signal Analysis

Programming: Python, SQL, C++, Julia, R

ML/DL Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, spaCy, NLTK, OpenCV

Tools: Jupyter Notebook, Google Colab, VS Code, PyCharm, Git, GitHub, GitLab, Docker, Flask

Databases & Modeling: SQL Server, Relational Database Design, ERD, SQL Programming

Visualization: Power BI, Tableau, Plotly/Dash

Cloud: Google Cloud Platform (GCP)

Workflow: Notion, Trello, Scrum

Languages

- **Persian:** Native
- **English:** Professional Working Proficiency | CEFR Level: B2
- **German:** Intermediate (A2-B1)

Honors & Achievements

- **Kaggle Master** (Sep 2021 – Nov 2025), highest notebook ranking: 70th place [\[Kaggle Profile\]](#)
- **Professional Activity:** Ranked **#3 in Data Science** (Iran) and recognized as “**The AI Luminary**” for impactful AI education and content creation on LinkedIn by Favikon (2025). [\[View Report\]](#)

Certifications

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| Comprehensive Data Science Program
<i>350-hour specialization (GPA: 19.44/20 (German Grade: 1.1))</i> [View Certificate] | 2025
<i>University of Tehran</i> |
| ● Courses: Fundamentals of Statistics & Linear Algebra, Data Analysis in Excel, Fundamentals of Data Science, Database Management in SQL Server, Data Science in Python, Statistical Analysis in R, Process Management and Process Mining, Data Storytelling in Tableau, Business Intelligence with Power BI, Practical Text Mining, Web Mining and Social Network Analysis, Deep Learning, Data Management based on DMBOK Framework, Big Data Analytics, Time Series with Python, Recommender Systems with Python, Special Issues. | |
| Comprehensive Data Science Professional Program
[View Certificate] | 2022
<i>Jahad Daneshgahi Sharif</i> |
| Advanced AI & Healthcare Workshops
<i>Specialized training tracks supervised by Dr. Zohreh Azimifar:</i> | 2021
<i>Shiraz University & SUMS</i> |
| ● Data Science & AI in Healthcare – [View Certificate] | |
| ● Introduction to Deep Learning – Supported by SUWFEA. [View Certificate] | |
| Advanced Academic Seminars | 2023 – 2024
<i>Sharif University of Technology (WSS)</i> |
| ● Workshop on Large Language Models – 9th WSS, 2024. [View Certificate] | |
| ● Advanced Topics in Computer Science and Engineering – 9th WSS, 2024. [View Certificate] | |
| ● Advanced Topics in Computer Science and Engineering – 8th WSS, 2023. [View Certificate] | |

References

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| Academic References:
<i>Undergraduate Thesis & Academic Instructors</i> | Sirjan University of Technology |
| ● Dr. Amir Salarpour – salarpour@sirjantech.ac.ir | [Recommendation Letter] |
| ● Dr. Ahmad Pouramini – pouramini@sirjantech.ac.ir | [Recommendation Letter] |
| ● Somayeh Khajehasani – khajeh@sirjantech.ac.ir | [Recommendation Letter] |
| Professional References
<i>Mentors & Supervisors:</i> | Industry & Research |
| ● Dr. Hossein Sameti – sameti@sharif.edu | [Recommendation Letter] |
| ● Pooria Haddad | [Work Certificate] |
| ● Mahmoud Alipour – Mahmoudalipour.a@gmail.com | [Recommendation Letter] |